

network, the rate of creation is kept static. So the more popular it gets, the harder it becomes to generate bitcoins. So much that normal computers can no longer generate bitcoins at a profit. The amount spent for the electricity will far outweigh the value of the generated bitcoins. Yet people continue to mine. This is because of the specialised hardware that has been released which can generate bitcoins at a pace far greater than ever before but while consuming far lesser amount of electricity..

Whenever anyone generates a proof-of-work it becomes part of the blockchain. The blockchain then serves as a ledger for every transaction that ever takes place. This is why all your transactions in the Bitcoin ecosystem can be traced from the very origin. Only after verification is the bitcoin transaction happen. So if anyone would've programmed a bitcoin wallet to show a balance of bitcoins that never existed it couldn't ever be used. This is called the double spending problem and the concept of bitcoin was conceived in order to solve the double spending problem. Whenever digital currency is generated, there is the possibility of copying that currency. And thus creating a duplicate which can be used like the original currency. Bitcoin does not have this issue thanks to the blockchain. Any person trying to scam the system needs to rewrite a good portion of the blockchain and that can only be done via a consensus. That is, a majority of the clients hosting the blockchain must be convinced that a new addition/modification is infact been seen by a majority of the clients. Unless this happens the new addition/modification is rejected. So it would take a few of the world's fastest supercomputers to actually change the balance of the bitcoin economy. All of this makes the Bitcoin system very difficult to scam.

Glossary

Usually reserved for the end, it would be advisable to go through the glossary now itself to get a better understanding of things to follow.

Bitcoins - It is a cryptocurrency that is generated and used within the bitcoin ecosystem. The currency is very volatile and the current value is roughly around USD 900.

Block - Blocks are individual links in a chain of transaction verifications. Transactions that are outstanding get added into a block and are verified approximately every ten minutes. Each new block generated helps with the verification of all previous blocks since it adds upon the previous one. Data is permanently recorded in the Bitcoin distributed network through files called blocks.